INCH-POUND
MIL-DTL-24643/35C
22 August 2002
SUPERSEDING
MIL-C-24643/35B
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3SU
(INCLUDING VARIATION LS3SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

This cable can be manufactured in three variations: LS3SU and LS3SUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, AWG Size No. 18 (7 by 0.0152 inch), uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.
Fourth -	Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth -	Binder tape over each triad, applied at manufacturer's option, applied helically with overlap.
Sixth -	Braided shield of AWG No. 34 or No. 36 bare copper. Braided angles of 30 to 35 degrees; minimum coverage of 85 percent.

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Construction (non-watertight) (continued)

Seventh - Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed, An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

Eighth - The required number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS3SU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

Construction variant for type LS3SUS

Tenth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Eleventh - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) Separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Thirteenth - Binder tape applied helically with overlap.

Fourteenth - Cross-linked polyolefin jacket. (See table I for thickness).

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TABLE I. Details.

Military part no. 1/ M24643/35	Size 2/	Number of triads	Cable jacket thickness min avg (inch)	Diameter over shielded core (inch nom.)	Overall diameter (inches)			
					TYPE LS3SU		TYPE LS3SUS	
					min	max	min	max
-01UN	3	3	0.040	0.559	0.650	0.700	0.710	0.760
-02UN	7	7	0.050	0.761	0.840	0.910	0.900	0.970
-03UN	10	10	0.050	0.987	1.100	1.190	1.160	1.250
-04UN	14	14	0.050	1.049	1.200	1.290	1.260	1.350
-05UN	19	19	0.050	1.199	1.340	1.430	1.400	1.490
-06UN	24	24	0.060	1.370	1.580	1.670	1.640	1.730
-07UN	30	30	0.060	1.460	1.680	1.770	1.740	1.830
-08UN	37	37	0.060	1.575	1.840	1.930	1.900	1.990
-09UN	44	44	0.060	1.800	2.060	2.150	2.120	2.210

1/ When double overall shield is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LS3SU and LS3SUS.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 7.06
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
  - Shield to shield (Triad shields only) 500
  - Shield to shield (LS3SUS overall shields only) 200
- Insulator resistance - Megohms/1000 feet minimum
  - Conductor to conductor 500
  - Component shield to shield 100
  - Shield to shield (LS3SUS overall shields only) 100
- Jacket flaws (LS3SUS only) ..... No failure
- Conductor and shield continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure
- Capacitance
  - Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum 30
  - Capacitance unbalance - percent maximum 10

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50

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Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage ..... No failure
Surface transfer impedance (Type LS3SUS only)
Milliohms per meter, maximum 700
EMP response time - dB, minimum 60

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Shield insulation 2

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Halogen content - percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-033)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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